



Vulnerability analyses of power-water interdependent infrastructure networks by using load-functional dependency coupled cascading failure model

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ABSTRACT

Previous studies on vulnerability analyses of interdependent critical infrastructure networks are mainly based on cascading failure models that only consider the effect of network topologies and binary dependencies, overlooking progressive functional degradation. This study proposes a novel cascading failure model that explicitly couples load dynamics with progressive functional dependency in power-water interdependent infrastructure networks. By introducing execution performance parameter as a quantitative indicator, the proposed model establishes a bidirectional feedback mechanism between load redistribution and functional dependency-induced performance loss. The proposed model is applied to analyze vulnerability of a county-scale power-water interdependent infrastructure network in Zibo, China. Results show that vulnerability of power-water interdependent infrastructure networks will be underestimated by neglecting dynamic coupling of load dynamics and functional dependency. Moreover, node attributes only account for 24.5% of the functional degradation, suggesting that network vulnerability predominantly emerges from dynamic coupling of load dynamics and functional dependency. This study provides a new practical approach for analyzing vulnerability and improving resilience of interdependent critical infrastructure networks.

1. Introduction

Urban infrastructures are fundamental systems that support the normal operation of cities and ensure public safety, and their stability is directly related to the continuity of socioeconomic activities and the emergency response capability of urban systems under extreme events [1]. With the continuous expansion of urban scale and the increasing operation complexity of infrastructure systems, the interconnections among different infrastructure systems have become increasingly pronounced, where localized disturbances in a single infrastructure system can be easily amplified into inter-system risks [2,3].

Among urban infrastructure systems, power and water infrastructures are typical interdependent critical infrastructures. Water systems rely heavily on power supply for water extraction, pressurization, transmission, distribution, and treatment, while power systems depend on stable water supply for power generation cooling, equipment operation, and emergency support [4–6]. Due to the interdependence between power and water infrastructures, localized failures in one

system under external disturbances may propagate across infrastructure networks through dependency pathways, thereby triggering cascading failures in interdependent power-water infrastructure networks [7–11]. During cascading failure processes, initial disturbances are progressively amplified, leading to the spatial and temporal spread of system functionality loss and posing severe threats to urban operational safety.

Extensive studies have been conducted on analyzing the vulnerability of interdependent infrastructure networks by using various cascading failure models, where interdependent infrastructure systems are modeled as multilayer networks with explicit interdependencies by using complex network theory [12–16]. Existing cascading failure models used in vulnerability analyses of power-water interdependent infrastructure networks can generally be categorized into three types [17–20]. The first type is load redistribution-based cascading failure models, which primarily consider cascading failures caused by load redistribution [21–24]. These models assume that when nodes in infrastructure networks fail, their loads will be redistributed to corresponding connected nodes, which may trigger cascading failures. Motter

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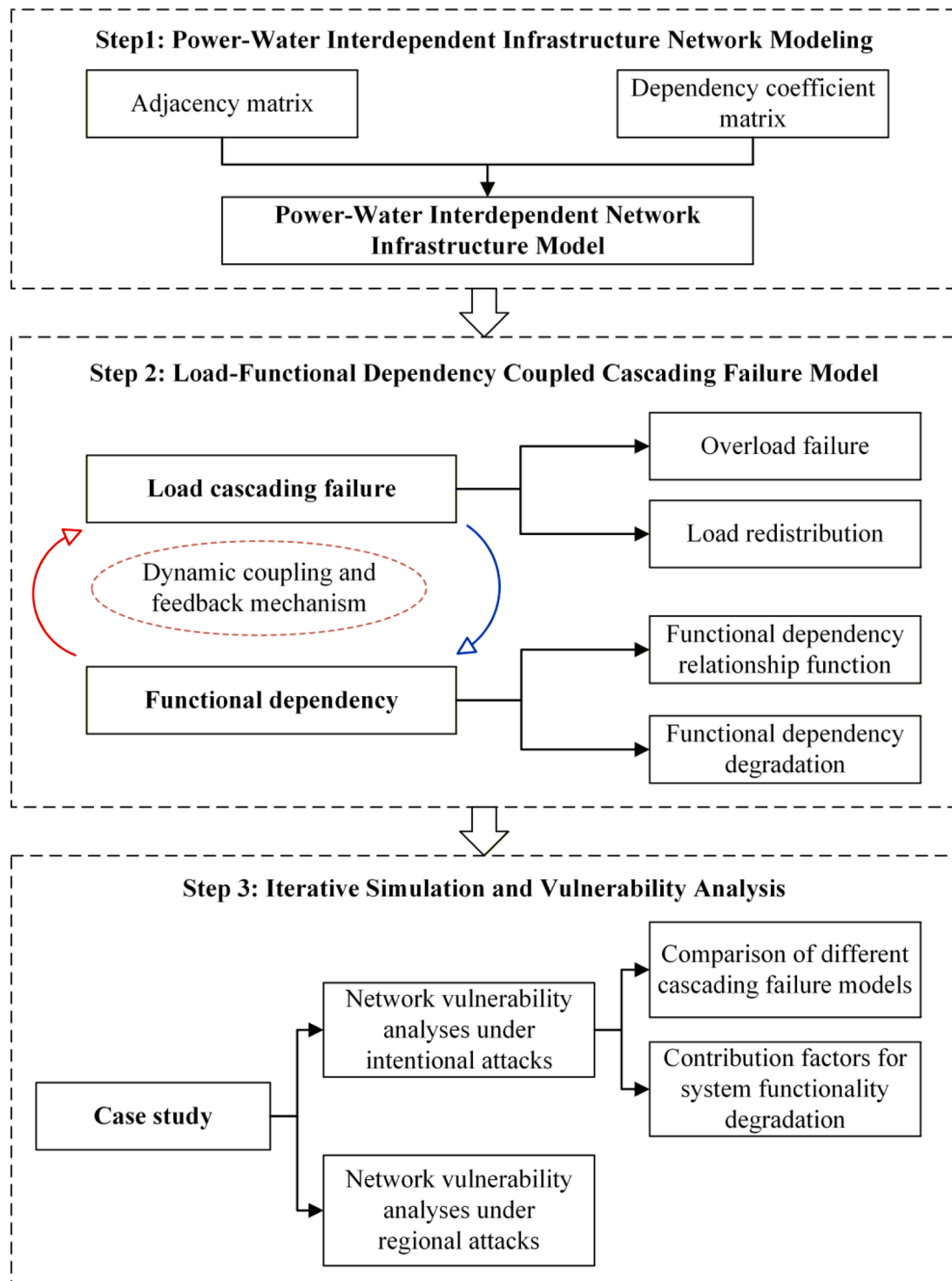


Fig. 1. Research framework of this study.

and Lai [21] proposed a classical load redistribution model to analyze cascade-based attacks in complex networks and demonstrated that overload propagation can lead to nonlinear vulnerability under targeted attacks. Fan et al. [23] further developed a cascading failure model for power grids based on complex network theory and founded that load redistribution significantly affects system robustness under disturbances. The second type is functional dependency cascading failure models, which mainly focus on cascading propagation caused by dependency loss across interdependent infrastructure systems [25–28]. In these models, the failure of one infrastructure component may directly

either reduce the functionality or cause failure of dependent components in another infrastructure network. Shao et al. [28] investigated cascading failures in coupled network systems with multiple support-dependence relations and revealed the critical role of dependency structures in failure propagation. Rueda and Calle [25] analyzed targeted attacks in interdependent power and telecommunication networks using dependency matrices and demonstrated the necessity of vulnerability assessment considering functional dependency in interdependent infrastructure networks. It should be noted that there is also a group of studies that consider functional dependency in

power-water interdependent infrastructure networks by modeling water infrastructure systems with power constraints [29–32]. For example, Zhang et al. [31] modeled a power-water interdependent infrastructure network by incorporating power constraints into water infrastructure operation and analyzed the impacts of power outages on water supply capacity. The third type models, which have been more recently proposed, integrate load redistribution cascading failure and functional dependency cascading failure mechanisms [33–35]. These models provide a more comprehensive representation of cascading failure propagation by considering both overload propagation and interdependency effects. For example, Yin et al. [34] proposed a hybrid cascading failure model incorporating fluctuating load and dependency clusters to evaluate network robustness. Fu et al. [35] developed a coupled cascading failure model with functional dependence in complex networks, where load redistribution and dependency propagation are modeled sequentially. Due to the improved capability in representing multiple failure mechanisms and better consistency with real-world infrastructure operations, cascading failure models that consider both overload propagation and interdependency effects are receiving increasing attention in the field of vulnerability analyses of interdependent infrastructure networks.

However, despite these advances, the previous cascading failure models for vulnerability analyses of interdependent infrastructure networks still have two major limitations. Firstly, load variation processes and functional dependency effects are usually modeled separately in the cascading failure models. These models often simulate load redistribution caused by node or link failures first, and then independently determine the failure of dependent nodes based on predefined functional dependency relationships [35–37]. Secondly, infrastructure nodes are commonly assumed to undergo instantaneous state transitions between normal operation and complete failure, causing cascading failure processes to exhibit pronounced binary failure characteristics. Although this assumption simplifies model formulation and computation, it limits the ability of the models to represent complex failure behaviors. Plenty of extreme natural disasters have demonstrated that the functional evolution of critical infrastructures during disasters usually is featured with progressive degradation rather than instantaneous complete failure [38,39]. Recent studies have attempted to move beyond binary-state assumptions by introducing partial functionality of disrupted infrastructure components or partial interdependence between infrastructure networks [40–42]. For example, Almoghatawi et al. [42] modeled post-disruption infrastructure components using partial functionality states to better capture post-disruption recovery processes in interdependent networks. However, these studies still do not explicitly consider the dynamic feedback between progressive functionality degradation and load redistribution in interdependent infrastructure networks.

To address the above-mentioned shortcomings, this study proposes a load-functional dependency coupled cascading failure modeling framework for power-water interdependent infrastructure networks. The main contributions and innovations of this study are summarized as follows:

- (1) A load-functional dependency coupled cascading failure model is proposed for vulnerability analysis of power-water interdependent infrastructure networks. The proposed model establishes an explicit bidirectional dynamic feedback mechanism between load redistribution and functional dependency-induced performance degradation, thereby mitigating the vulnerability underestimation caused by traditional cascading failure models that neglect such dynamic coupling.
- (2) The proposed model incorporates progressive functional degradation into the cascading failure process by introducing execution performance parameter and dependency coefficient. Compared with traditional binary failure assumptions, the proposed method can better characterize the nonlinear and multi-

stage propagation characteristics of cascading failures in power-water interdependent infrastructure networks.

The rest of this paper is organized as follows. Section 2 presents the overall methodology of this study, including the construction of the power-water interdependent infrastructure network and the development of the load-functional dependency coupled cascading failure model. Section 3 presents validation and discussion through vulnerability analyses of a county-scale power-water interdependent infrastructure network in Zibo, China. Section 4 summarizes the main conclusions of this study and discusses the limitations of the proposed model as well as future research directions.

2. Methodology

The overall research framework of this study is illustrated in Fig. 1, which consists of three main steps: (1) constructing power-water interdependent infrastructure networks that simultaneously consider intra-layer and inter-layer dependency relationships; (2) developing the load-functional dependency coupled cascading failure model; (3) iterative simulation and vulnerability analyses.

2.1. Power-water interdependent infrastructure network considering intra-layer and inter-layer dependencies

Consider a power-water interdependent infrastructure network G^{p-w} which consists of two network layers, i.e., power infrastructure network G^p and water infrastructure network layer G^w . Power infrastructure network G^p contains power infrastructure nodes such as power plants and substations. Water infrastructure network G^w contains water infrastructure nodes such as water plants, pump stations, water distribution plants, etc.

The node set V^p in the power infrastructure network layer and V^w in the water infrastructure network layer can be denoted as:

$$\begin{cases} V^p = \{v_1^p, v_2^p, \dots, v_{n_p}^p\} \\ V^w = \{v_1^w, v_2^w, \dots, v_{n_w}^w\} \end{cases}, \quad (1)$$

where n_p is the number of power infrastructure nodes in G^p and n_w is the number of water infrastructure nodes in G^w .

There are two types of dependency relationships, i.e., intra-layer dependency and inter-layer dependency. Taking intra-layer dependency in power infrastructures as example, 500 kV substations have dependency on power plants for original electricity supply while 220 kV substations have dependency on 500 kV substations for high-voltage electricity supply. Moreover, significant inter-layer dependencies exist between power and water infrastructures. On one hand, water infrastructures have dependency on electricity for pumping, pressurization, and distribution processes. On the other hand, power infrastructures have dependency on stable water supply for power generation cooling, equipment operation, and emergency support. For simplicity, only the unidirectional inter-layer dependency of water infrastructures on power infrastructures is considered in this study.

Dependency relationships between nodes can be represented by directed edges $v_j \rightarrow v_i$, indicating that node v_j provides support to node v_i . Thus, node v_i is the dependent node, and node v_j is the supporting node.

Adjacency matrix A represents the existence of directed connections between nodes within the same network layer, which can be expressed as $A = [a_{ij}]$. a_{ij} is 1 if dependent node v_i depends on supporting node v_j , otherwise a_{ij} is 0.

Traditional interdependent infrastructure network models usually use adjacency matrices or binary dependency relationships to describe whether dependency exists between nodes, while the strength of dependency between infrastructure nodes is not explicitly quantified.

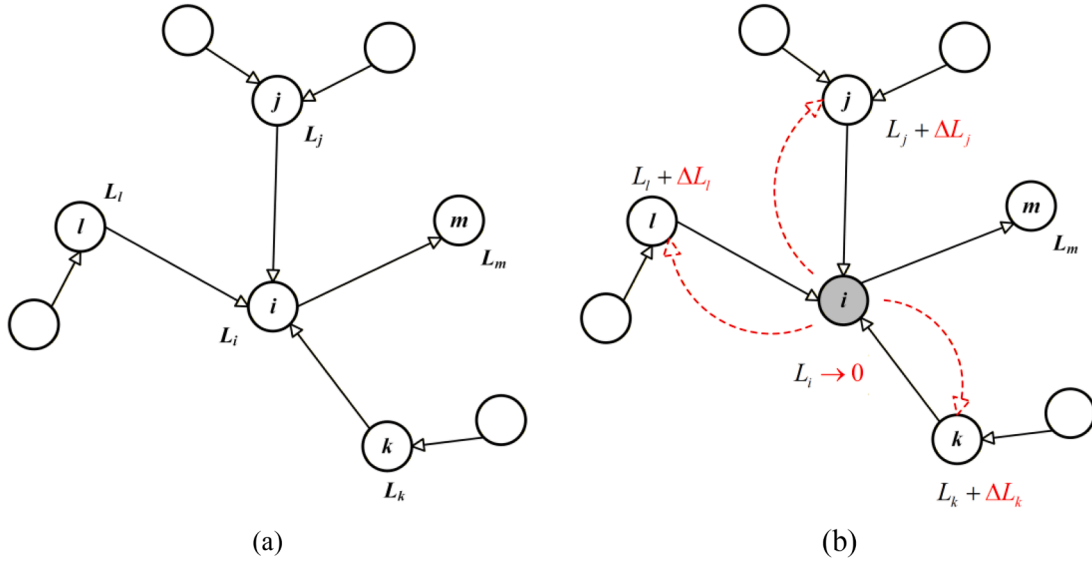


Fig. 2. (a) Initial node loads and (b) node loads after redistribution caused by a failure node.

Therefore, it is difficult to characterize heterogeneous functional dependency relationships among infrastructure nodes. Different from adjacency matrix A , dependency coefficient matrix $\Psi = [\varphi_{ij}]$ can not only reflect intra-layer dependency but also inter-layer dependency. Dependency coefficient $\varphi_{ij} \in [0, 1]$ quantifies the functional dependency strength of dependent node v_i on its supporting node v_j . Dependency coefficients can be determined based on power supply capacity, supply reliability, geographical proximity, and voltage level matching. The larger the dependency coefficient φ_{ij} is, the stronger the dependence of the dependent node on its supporting node is. Then, the power-water interdependent infrastructure network G^{p-w} can be constructed by the dependency coefficient matrix, where φ_{ij} with non-zero value corresponds to a directed edge $v_j \rightarrow v_i$.

Constraint coefficient matrix of the power-water interdependent infrastructure network can be written as $\Phi = [\phi_{ij}]$, where constraint coefficient $\phi_{ij} \in [0, 1]$ quantifies the constraint strength of supporting node v_j on dependent node v_i . The smaller the constraint coefficient ϕ_{ij} is, the higher the reliance of dependent node v_i on supporting node v_j is.

Independent operating capability $\Delta_i \in [0, 1]$ of node v_i is introduced to describe the ability where node v_i can maintain operation itself in the absence of external support from its supporting nodes. The larger the independent operating capability of a node is, the stronger the autonomous operational capability of the node is.

Assuming that node v_i has n supporting nodes, the relationship of dependency coefficient φ_{ij} , constraint coefficient ϕ_{ij} and independent operating capability Δ_i can be given as:

$$\begin{cases} \sum_{j=1}^n \varphi_{ij} + \Delta_i = 1 \\ \phi_{ij} = \sum_{k=1, k \neq j}^n \varphi_{ik} + \Delta_i \end{cases} \quad (2)$$

It can be easily seen from Eq. (2) that as the number of supporting nodes of node v_i increases, the impact of arbitrary supporting node's failure on node v_i becomes diluted, thereby enhancing the overall robustness of the interdependent infrastructure network.

2.2. Load-functional dependency coupled cascading failure model

To characterize the dynamic evolution of power-water interdependent infrastructure networks under disturbances, load cascading failure and functional dependency cascading failure are coupled within a

Algorithm 1

The algorithm for node loads redistribution in power-water interdependent infrastructure networks.

Function: Redistribute node loads in the corresponding systems of the failure nodes
Input: Interdependent infrastructure network (G^{p-w}), Adjacency matrix (A), Set of failure nodes (FNS), Node loads (NL), Regular remaining capacity of nodes (RRC), Ultimate remaining capacity of nodes (URC)
Output: Node loads after load redistribution (NLR)

```

1: for each failure node  $v_i$  in  $FNS$ :
2:   Retrieve the support nodes of node  $v_i$  in the corresponding system whose
   adjacency coefficient  $A_{ij}$  is not 0  $\rightarrow SN_i$ 
3:   Retrieve the nodes whose  $RRC$  is larger than 0 from  $SN_i \rightarrow SNR_i$ 
4:   if  $SNR_i$  is not null:
5:     if the addition of  $RRC$ s of the  $SN_i \geq$  the load of the failure node  $NL_i$ :
6:       Redistribute  $NL_i$  among  $SNR_i$  in proportion to their  $RRC$ 
7:     else:
8:       Redistribute partial  $NL_i$  among  $SNR_i$  in proportion to their  $RRC$ s until
       they become 0
9:   Update the  $RRC$ s and  $URC$ s of the  $SN_i$ 
10:  if the addition of  $URC$ s of the  $SN_i$  is not 0:
11:    Redistribute the rest of  $NL_i$  among  $SN_i$  in proportion to their  $URC$ s
12:  else:
13:    Add zero loads to  $SN_i$ 
14:  else:
15:    if the addition of  $URC$ s of the  $SN_i$  is not 0:
16:      Redistribute the rest of  $NL_i$  among  $SN_i$  in proportion to their  $URC$ s
17:    else:
18:      Add zero loads to  $SN_i$ 
19: return  $NLR \rightarrow$  update  $NL$  where  $NL_i$  is 0

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unified framework.

2.2.1. Load cascading failure

In power-water interdependent infrastructure networks, the operational status of a node is constrained by its load-capacity relationship. Let node load L_i represent the workload borne by node v_i . Capacities of node v_i can be divided into regular capacity C_i^{reg} and ultimate capacity C_i^{max} , corresponding to the maximum load that node v_i can sustain without efficiency loss and the maximum load it can sustain before failure, respectively.

Efficiency f_i of node v_i can reflect its load-processing capability under current performance conditions. When node load is less than regular capacity, node efficiency remains equal to 1. When node load is between the regular capacity and the ultimate capacity, node efficiency decreases linearly as node load increases. When node load exceeds ultimate capacity, this node fails and its efficiency drops to 0. Hence, efficiency f_i of

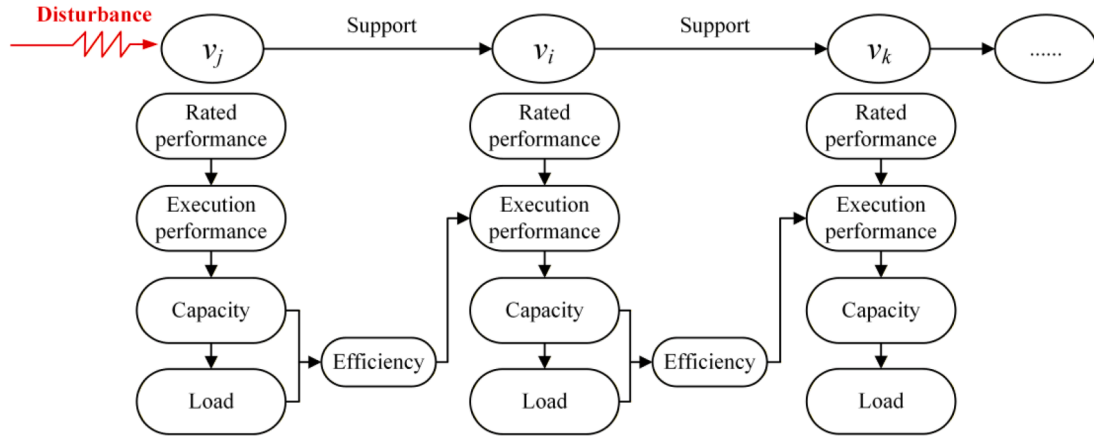


Fig. 3. Schematic of interaction in load-capacity-efficiency-performance.

node v_i can be given as:

$$f_i = \begin{cases} 1, L_i \leq C_i^{\text{reg}} \\ \frac{C_i^{\text{max}} - L_i}{C_i^{\text{max}} - C_i^{\text{reg}}}, C_i^{\text{reg}} < L_i < C_i^{\text{max}} \\ 0, L_i \geq C_i^{\text{max}} \end{cases} \quad (3)$$

When efficiency of a node decreases to 0, its load will be redistributed to its supporting nodes within the same network layer, as illustrated in Fig. 2. Load redistribution follows the principle of remaining capacity priority [23], and the detailed procedure is given in Algorithm 1, which can be described as follows:

Define regular remaining capacity (RRC_i) of node v_i as $C_i^{\text{reg}} - L_i$, ultimate remaining capacity (URC_i) of node v_i as $C_i^{\text{max}} - L_i$. In the case that all supporting nodes of the failure node satisfy $RRC_j > 0$ (i.e., all supporting nodes are non-overloaded), load of the failure node will be redistributed proportionally according to their regular remaining capacities RRC_j . In the case that all supporting nodes of the failure node are overloaded (i.e., $RRC_j = 0$ and $URC_j > 0$), load of the failure node will be redistributed proportionally according to their ultimate remaining capacities URC_j . In the case that both non-overloaded and overloaded supporting nodes exist, if the sum of regular remaining capacities RRC_j of non-overloaded supporting nodes are insufficient to absorb load of the failure node, partial load will be firstly allocated to the non-overloaded nodes according to their regular remaining capacities RRC_j . Then, the remaining load of the failure node will further be redistributed to all the overloaded supporting nodes, including those newly get overloaded by absorbing the partial redistributed load, according to their ultimate remaining capacities URC_j .

2.2.2. Functional dependency

To describe functional dependency effects of dependent nodes on supporting nodes, two performance parameters are introduced: rated performance parameter $Q_i \in [0, 1]$ and execution performance parameter $p_i \in [0, 1]$. Rated performance parameter Q_i reflects the intrinsic functionality and its initial value is 1. Under disturbances, Q_i will decrease gradually from 1 to 0. Execution performance parameter p_i of node v_i represents the node's ability to realize its rated performance parameter Q_i under current operating conditions.

According to the functional dependency network analysis method (FDNA) [43–45], with Eq. (2), supporting strength of the operational capability of node v_i from supporting node v_j SOD_{ij} can be written as:

$$SOD_{ij} = \varphi_{ij} f_j + (1 - \varphi_{ij}). \quad (4)$$

Constraint strength of the operational capability of node v_i from supporting node v_j COD_{ij} can be written as:

$$COD_{ij} = f_j + \phi_{ij}. \quad (5)$$

The supporting strength SOD_{ij} and constraint strength COD_{ij} of node v_i can be expressed as:

$$\begin{cases} SOD_i = \text{Average}(SOD_{i1}, SOD_{i2}, \dots, SOD_{in}) \\ COD_i = \text{Min}(COD_{i1}, COD_{i2}, \dots, COD_{in}) \end{cases} \quad (6)$$

where SOD_i denotes the averaged supporting strength from all the supporting nodes of node v_i and COD_i denotes the minimal constraint strength effects of all the supporting nodes on node v_i .

Thus, the execution performance parameter p_i of node v_i can be obtained by:

$$p_i = \min(SOD_i, COD_i). \quad (7)$$

As the number of supporting nodes increases, the average effect of SOD_i tends to smooth out, resulting in a diminishing marginal improvement, whereas the constraint impact of COD_i becomes more pronounced.

2.2.3. The dynamic coupling and feedback mechanism of load failure and functional dependency

A degradation in the efficiency of a supporting node propagates to its dependent node through functional dependency, leading to a reduction in the dependent node's capacity. Capacity reduction of the dependent node further affects its efficiency and triggers load redistribution, thereby forming a loop coupling among load dynamics, functional performance, capacity, and efficiency.

Fig. 3 illustrates the interaction mechanism of load, capacity, efficiency, and execution performance parameter in the coupled cascading failure process. Specifically, the efficiency f_j of a supporting node v_j influences the execution performance parameter p_i of its dependent node v_i , which in turn determines the capacity of node v_i . The resulting capacity variation changes the efficiency f_i , and the updated efficiency f_i further propagates through functional dependency relationships to subsequent dependent nodes, forming the cascading process throughout the entire infrastructure network [35].

According to Fig. 3, node capacity is jointly determined by execution performance parameter and rated performance parameter. It is assumed that node capacity decreases proportionally with reductions in execution performance parameter and rated performance parameter, which can be expressed as:

$$\begin{cases} EC_i^{\text{reg}} = Q_i \cdot p_i \cdot C_i^{\text{reg}} \\ EC_i^{\text{max}} = Q_i \cdot p_i \cdot C_i^{\text{max}} \end{cases} \quad (8)$$

Accordingly, the node efficiency function given by Eq. (3) is modified as follows:

Algorithm 2

The algorithm for node efficiency computation in power-water interdependent infrastructure networks considering functional dependency.

Function: Calculate node efficiency in the infrastructure networks considering functional dependency

Input: Interdependent infrastructure network (G^{p-w}), Dependency coefficient matrix (Ψ), Set of failure nodes (FNS), Set of low efficiency nodes ($LENS$), Node loads (NL), Execution regular capacity (EC^{reg}), Execution ultimate capacity (EC^{max})

Output: Node efficiencies (f)

```

1:   for each node  $v_j$  in  $FNS$  and  $LENS$ :
2:     Retrieve the affected nodes of the node  $v_j$  in the  $G^{p-w}$  whose dependency
       coefficient  $\varphi_{ij}$ 
       is not 0  $\rightarrow AN_j$ 
3:     if  $AN_j$  is not null:
4:       for each node  $v_i$  in  $AN_j$ :
5:         if the node  $v_i$  is already in  $FNS$ :
6:           continue
7:         else:
8:           Retrieve the support nodes of the node  $v_i$  whose dependency coefficient  $\varphi_{ij}$ 
             is not 0  $\rightarrow SN_i$ 
9:           Calculate the average value of  $SODs$  of nodes in  $SN_i$  via Eq. (4)  $\rightarrow SOD_i$ 
10:          Calculate the minimal value of  $CODs$  of nodes in  $SN_i$  via Eq. (5)  $\rightarrow COD_i$ 
11:          Determine the execution performance parameter  $p$  of the node  $v_i$  via Eq.
             (7)  $\rightarrow p_i$ 
12:          Calculate  $EC_i^{reg}$ ,  $EC_i^{max}$  and  $f_i$  via Eq. (8–9)
13:        else:
14:          continue

```

$$f_i = \begin{cases} 1, L_i \leq EC_i^{reg} \\ \frac{EC_i^{max} - L_i}{EC_i^{max} - EC_i^{reg}}, EC_i^{reg} < L_i < EC_i^{max} \\ 0, L_i \geq EC_i^{max} \end{cases} \quad (9)$$

Algorithm 2 summarizes the updating procedure of node execution performance parameter and node efficiency under the load-functional dependency coupled cascading mechanism. Owing to the interaction among load redistribution, capacity degradation, and functional dependency propagation, the cascading failure process exhibits nonlinear propagation characteristics and progressive functional degradation.

2.3. Simulation procedure of the coupled cascading failure model

2.3.1. Attack strategy

All nodes in the power-water interdependent infrastructure network

are assumed to operate under normal conditions, with node loads, capacities, rated performance parameters, execution performance parameters, and efficiencies set to their initial values. The simulation procedure begins with the specification of disturbance inputs, including attack strategies and corresponding attack nodes. To examine vulnerability characteristics of power-water interdependent infrastructure networks under different disturbance scenarios, two typical attack strategies are considered: intentional attack strategy and regional attack strategy.

(1) Intentional attack strategy

Intentional attack strategy is designed to simulate intensive disturbance to one targeted node under one attack. Thus, in the scenario of intentional attacks, the node under attack will be treated as a failure node. Degree centrality, betweenness centrality and dependency degree of nodes have been widely used to analyze topologies of critical infrastructure networks [35,46,47]. Degree centrality of a node measures the number of direct dependency links associated with the node in the network. Betweenness centrality of a node can characterize its importance as transmission intermediary. Dependency degree of a node represents the number of both the direct and indirect dependent nodes supporting by this node. Moreover, node load can reflect the importance of a node in providing relevant services. Therefore, when generating the intentional attack node sequences, node sequences according to ranking of initial load, degree centrality, betweenness centrality, and dependency degree will be considered.

(2) Regional attack strategy

Regional attack strategy is designed to simulate regional disturbances to a group of nodes under one attack, which reflects disturbance scenarios of large-scale hazards such as earthquakes and floods. In this study, the concept of attack range (AR) is introduced to refer to total number of attack nodes in the regional attack strategy. After selecting a central attack node, breadth first search algorithm is used to obtain dependent nodes of the central attack node with different reachability distances. If the number of nodes below a certain reachability distance from the central attack node, including the central attack node itself, is enough to reach AR, breadth first search will terminate; otherwise, nodes with higher certain reachability distance have to be searched further. It should be noted that for a given attack range, there may be

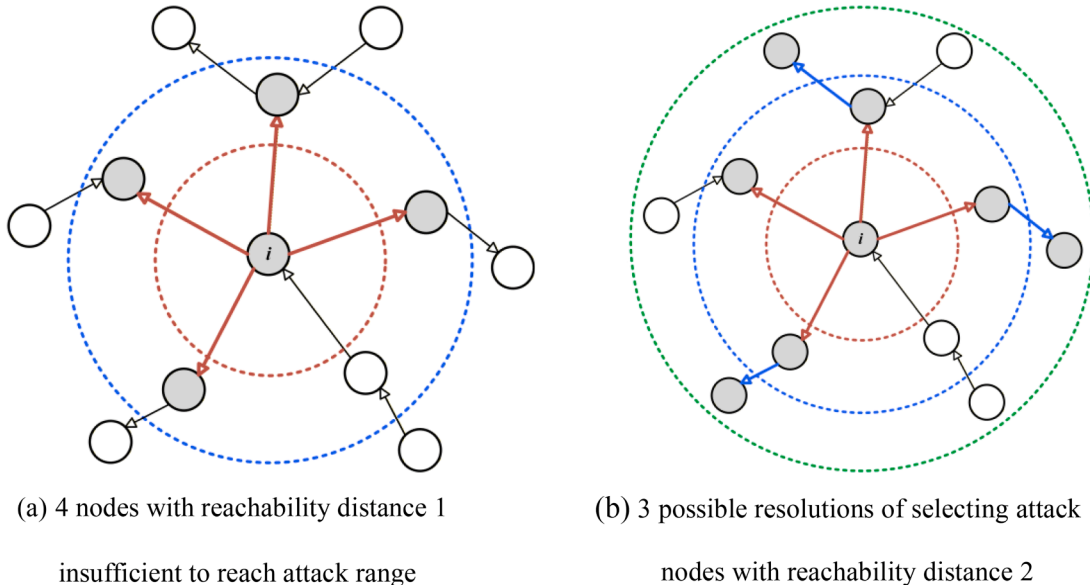


Fig. 4. Illustration of determination of attack nodes under regional attack.

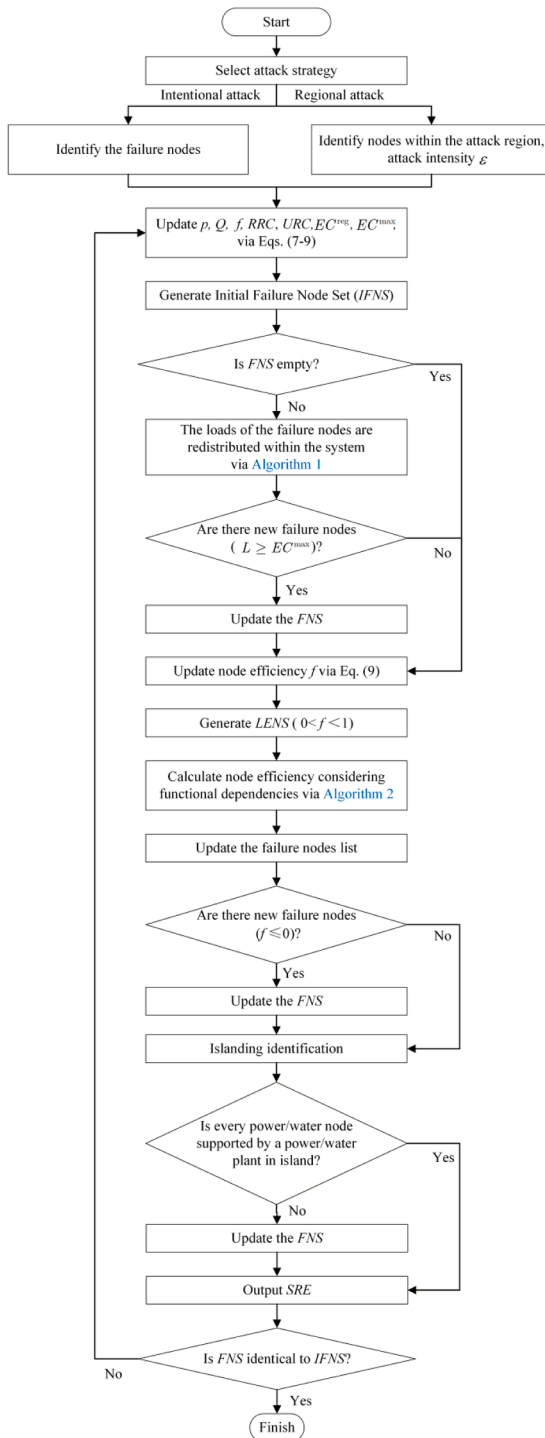


Fig. 5. Simulation procedure of the coupled cascading failure model.

different resolutions of attack nodes, as shown in Fig. 4. In this study, all different resolutions of attack nodes are computed for a given attack range, and the average value of these resolutions is assigned to the result for the given attack range. Attack intensity parameter $\varepsilon \in [0, 1]$ is then applied to reduce the initial rated performance parameter Q of nodes within the attack range, representing different levels of regional disturbance severity.

2.3.2. Simulation iteration

The coupled cascading failure model simulates the dynamic evolution of node states through an iterative updating process. The overall

simulation procedure is illustrated in Fig. 5. In each iteration, node states are updated sequentially according to the following steps.

(1) Execution of initial attack

Once the attack strategies and corresponding attack nodes are selected, state variables of the attack nodes, including execution performance parameter p , rated performance parameter Q , efficiency f , $EC^{\max}$, regular remaining capacity (RRC), and ultimate remaining capacity (URC), are updated according to Eqs. (7)–(9). Nodes whose efficiency decreases to 0 are identified as failure nodes and removed from the network topology, forming the initial failure node set ($IFNS$).

(2) Load redistribution

The loads of failure nodes are redistributed to their supporting nodes via Algorithm 1. After redistribution, node loads are updated and new failure nodes can be identified and added to the failure node set FNS .

(3) Efficiency updating considering functional dependency

Then, node efficiencies f_i are recalculated with the aid of Eq. (9). Nodes whose efficiencies decrease but do not reach complete failure (i.e. $0 < f_i < 1$), are classified into the low-efficiency node set ($LENS$). Using the dependency coefficient matrix Ψ and the constraint coefficient matrix Φ , functional dependency relationships among nodes can be identified. Algorithm 2 is then applied to the nodes affected by those in FNS and $LENS$, updating their execution performance parameter, capacities, and efficiencies, and identifying any new failure nodes to update the FNS .

(4) Islanding identification

After each iteration round of load redistribution and efficiency update considering functional dependency, island identification is performed to determine whether disconnected subgraphs exist [48]. In a subgraph in power infrastructure network or water infrastructure network, nodes without direct nor indirect support from power plant or water plant are treated as failure nodes and added to the FNS .

Then, system remaining efficiency (SRE) in this iteration round $SRE = \frac{1}{n_p + n_w} \sum_{i=1}^{n_p + n_w} f_i$ is outputted for subsequent vulnerability analyses. After that, the simulation procedure will make a decision on whether to continue. If the current FNS is not identical to the $IFNS$, which means the system has not reached a steady state yet, iterative process will move to the next iteration round. Otherwise, the iterative process terminates if the FNS remains unchanged over two consecutive iterations, indicating that the system has reached a steady state.

2.3.3. Time complexity of the simulation procedure

In every iteration round of the simulation procedure, computational costs are mainly concentrated in the module of load redistribution and that of efficiency updating considering functional dependency. The module of load redistribution involves one main for-loop to vary every failure node (see the first row in Algorithm 1), and computational cost in every loop is mainly concentrated in the case that both non-overloaded and overloaded supporting nodes exist in redistributing the load of the failure node, where involves an implicit inner for-loop to vary every non-overloaded supporting node to calculate its redistributed load. When programming the simulation procedure with coding languages supporting matrix computation, computation process in the inner for-loop can be operated via row-wise multiplication of the adjacency matrix whose row size is $(n_p + n_w)$. Thus, the worst-case time complexity of module of load redistribution is $O((n_p + n_w)^3)$. The module of efficiency updating considering functional dependency involves a double for-loop (see the first row and forth row in Algorithm 2). Similarly, the worst-case

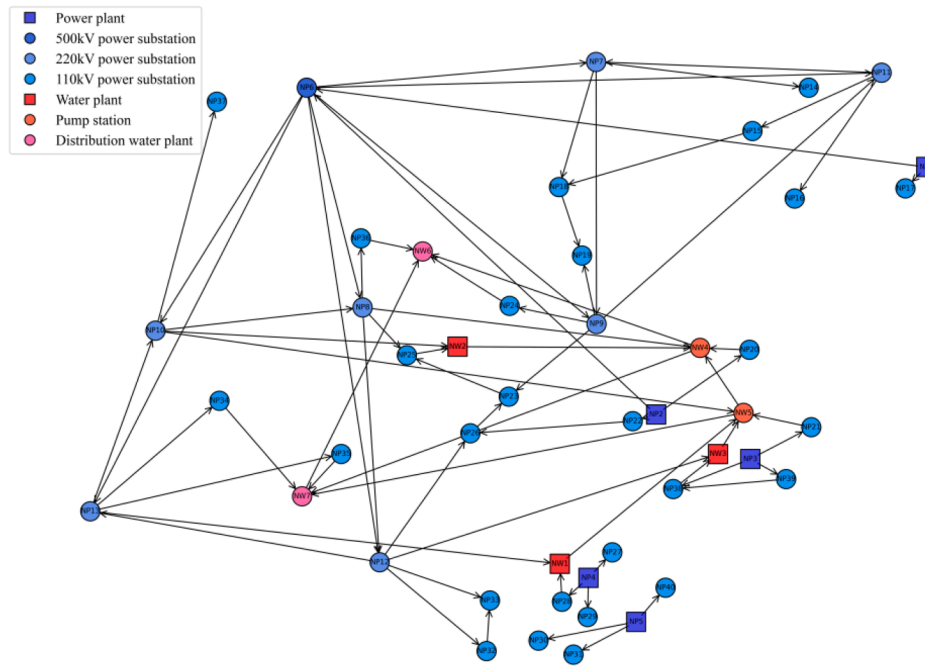


Fig. 6. A county-scale power-water interdependent infrastructure network in Zibo, China.

time complexity of module of efficiency updating considering functional dependency is also $O((n_p + n_w)^3)$. Thus, the worst-case time complexity of single iteration round of the simulation procedure is $O((n_p + n_w)^3)$, and the worst-case time complexity of the whole simulation procedure is $O((n_p + n_w)^4)$.

3. Validation and discussion

To validate the proposed load-functional dependency coupled cascading failure model, it is applied to analyze vulnerability of a county-scale power-water interdependent infrastructure network in Zibo, China, as shown in Fig. 6. There are 40 power infrastructure nodes and 7 water infrastructure nodes in the network, which provides services for 806 thousand people across 244 square kilometers of land. For power nodes, the initial load is characterized by apparent power, and node capacity is defined by rated power supply capability. For water nodes, the initial load is represented by water flow rate per unit time and capacity is defined by maximum water supply capability. Node attribute data are obtained from statistical records and engineering operational parameters of regional power and water facilities. Main node attributes of the power-water interdependent infrastructure network in the case study are listed in Table 1. Due to space limitation of this paper, other parameters, including adjacent matrix, dependency coefficient matrix and constraint coefficient matrix are included in the supplementary material of this paper.

3.1. Network vulnerability analyses under intentional attacks

3.1.1. Comparative analyses of different cascading failure models

The models are programmed in Python 3.11 and implemented in PyCharm, an integrated development environment (IDE) for Python, on a PC with an Intel Core i5-8265 U CPU at 1.80 GHz and 8 GB RAM. Average value of system remaining efficiencies in every simulation iteration of every individual node from all the 47 nodes under intentional attack obtained by the load-functional dependency coupled cascading failure model is compared with the results by other three traditional models, i.e., the functional dependency cascading failure model, the load cascading failure model, and the load and functional dependency cascading failure model in series where load cascading

failure propagation is first performed until the system reaches a steady state and then the functional dependency cascading failure process is subsequently implemented, as shown in Fig. 7. It should be noted that the final simulation iteration round where the iterative process terminates for different attack nodes varies, and the maximum iteration round in the computation in this study is set as 10. To ensure the results for different attack nodes by different models comparable, for attack nodes whose iterative process terminates in iteration round < 10 , it is assumed that remaining efficiency of the system keeps unchanged in the next several virtual iteration round until the virtual iteration round reaches 10. It should be noted that with the above-mentioned computing device, the average execution time of the simulation procedure of one node under intentional attack is about 1 s.

It can be found from Fig. 7 that system functionality decreases with cascading failure in all the four models. However, the extent of functionality degradation and the final steady-state levels in these four models differ remarkably. The coupled cascading failure model results in the most severe degradation of functionality, followed by the load and functional dependency cascading failure model in series, the functional dependency cascading failure model and the load cascading failure model. Moreover, *SRE* by the coupled cascading failure model exhibits a more rapid decline during the early stages of iterations. Compared with the functional dependency cascading failure model, the load and functional dependency cascading failure model in series leads to lower *SRE* values and more severe functionality degradation, indicating that the prior load redistribution process further aggravates subsequent dependency-induced cascading failures. Nevertheless, the degradation extent in the series model is still weaker than that in the coupled cascading failure model, since the interactions between load redistribution and functional dependency are considered sequentially rather than dynamically coupled during each iteration process. These results indicate that neglecting the coupling of load dynamics and functional dependency will underestimate system vulnerability in power-water interdependent infrastructure networks.

3.1.2. Analyses of contribution factors for system functionality degradation

To examine the effects of different node attributes on system functionality degradation, intentional attacks are successively applied to nodes ranked by initial load, degree centrality, betweenness centrality,

Table 1

Node attributes for the power-water interdependent infrastructure network in the case study.

Nodes	Load	Regular capacity	Ultimate capacity	Degree centrality	Betweenness centrality	Dependency degree
NP1	500	600	700	2	0.000	31
NP2	500	600	700	3	0.000	32
NP3	500	600	700	3	0.000	13
NP4	500	600	700	3	0.000	15
NP5	500	600	700	3	0.000	15
NP6	2572	3086	3601	9	0.026	29
NP7	386	463	540	5	0.009	15
NP8	386	463	540	6	0.015	20
NP9	386	463	540	6	0.013	15
NP10	386	463	540	6	0.013	20
NP11	386	463	540	5	0.006	15
NP12	386	463	540	7	0.018	20
NP13	386	463	540	6	0.013	20
NP14	107	128	150	1	0.000	0
NP15	107	128	150	2	0.001	2
NP16	107	128	150	1	0.000	0
NP17	107	128	150	1	0.000	0
NP18	107	128	150	3	0.001	1
NP19	107	128	150	2	0.000	0
NP20	107	128	150	2	0.001	3
NP21	107	128	150	2	0.002	4
NP22	107	128	150	3	0.014	7
NP23	107	128	150	3	0.018	5
NP24	107	128	150	2	0.002	1
NP25	107	128	150	3	0.014	4
NP26	107	128	150	3	0.016	6
NP27	107	128	150	3	0.014	11
NP28	107	128	150	2	0.002	5
NP29	107	128	150	1	0.000	0
NP30	107	128	150	1	0.000	0
NP31	107	128	150	1	0.000	0
NP32	107	128	150	2	0.000	1
NP33	107	128	150	2	0.000	0
NP34	107	128	150	2	0.001	2
NP35	107	128	150	2	0.001	2
NP36	107	128	150	2	0.001	1
NP37	107	128	150	1	0.000	0
NP38	107	128	150	5	0.025	10
NP39	107	128	150	2	0.000	11
NP40	107	128	150	2	0.006	12
NW1	5500	6600	7700	3	0.004	4
NW2	5500	6600	7700	3	0.009	3
NW3	5500	6600	7700	3	0.012	4
NW4	8300	9960	11,620	6	0.013	2
NW5	8300	9960	11,620	6	0.018	3
NW6	8300	9960	11,620	4	0.000	0
NW7	8300	9960	11,620	5	0.005	1

and dependency degree. The corresponding *SRE* under different attack sequences are shown in Fig. 8. It can be seen from Fig. 8 that under all four ranking-based attack sequences, *SRE* does not decrease monotonically as attacks proceed. Similar fluctuation patterns of *SRE* are observed for different node attributes.

To further quantify the contributions of node attributes on system functionality degradation, Spearman rank correlation analysis is conducted between each node attribute and *SRE*. The correlation coefficients and corresponding significance levels are summarized in Table 2. It can be seen from Table 2 that dependency degree has the largest negative correlation with *SRE* ($\rho = -0.477, p \leq 0.01$), followed by degree centrality ($\rho = -0.368, p \leq 0.05$). In contrast, the correlations between *SRE* and initial load as well as betweenness centrality are relatively weak and do not pass the significance test.

To further illustrate the correlation structure among node attributes and *SRE*, the Spearman correlation matrix is visualized in Fig. 9. It can be found from Fig. 9 that strong positive correlations exist among different node attributes, while the correlations between most individual attributes and *SRE* are comparatively weaker.

To evaluate the combined effects of initial network characteristics on system functionality degradation, a multiple linear regression analysis is conducted with *SRE* as the dependent variable. Normalized initial load,

degree centrality, betweenness centrality, and dependency degree whose minimal value is 0 and maximal value is 1 are included as independent variables. The regression results are summarized in Table 3. It is evident from Table 3 that the coefficient of determination of the regression model is $R^2 = 0.245$, indicating that the node attributes can only explain 24.5% of the variation of *SRE*. Moreover, the regression coefficients of initial load, degree centrality, betweenness centrality, and dependency degree do not pass the significance test at conventional significance levels.

The low explanatory power of the regression model suggests that a large proportion of variation of *SRE* cannot be attributed to node attributes, including initial load, degree centrality, betweenness centrality, and dependency degree. This indicates that system-level functionality degradation is mainly driven by dynamic coupling processes, including load redistribution and functional dependency evolution during cascading failures.

3.1.3. Cascading failure propagation characteristics of an ordinary node

To examine the coupling cascading failure propagation process, the evolution of cascading failures in the power-water interdependent infrastructure network with NP16 as intentional attack node is shown in Fig. 10. The reason why NP16 is chosen as an example is that the node

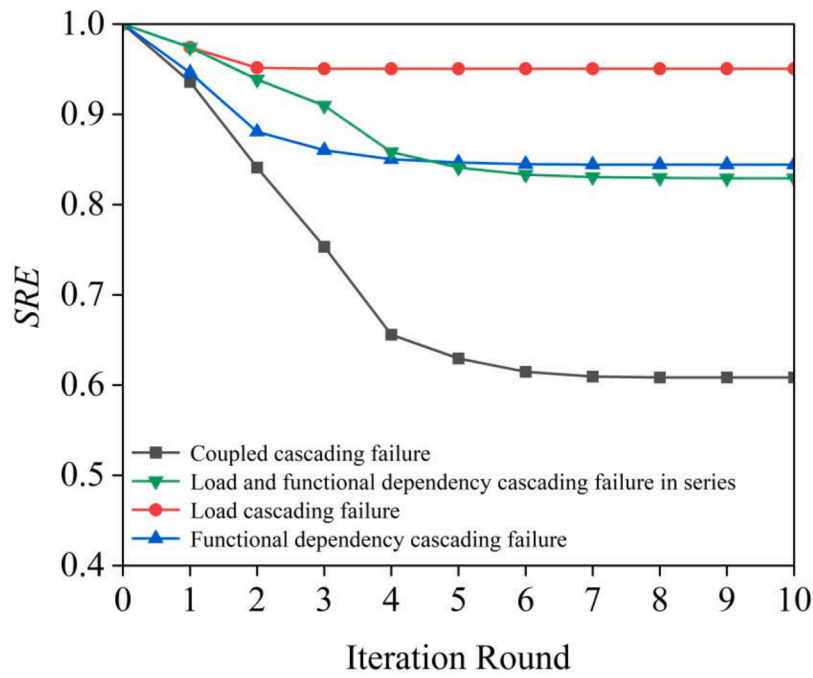


Fig. 7. Average value of SRE in every simulation iteration of all 47 nodes under intentional attack.

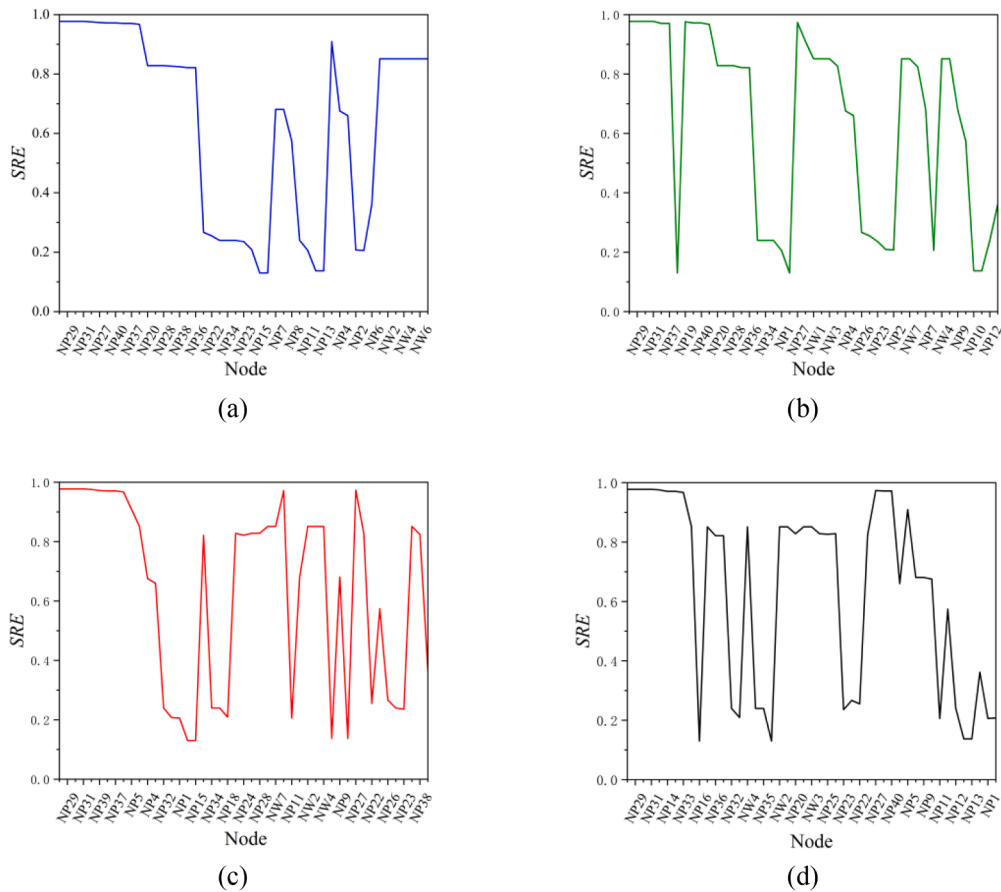


Fig. 8. SRE with attack node sequences according to ranking of (a) Load, (b) Degree centrality, (c) Betweenness centrality, (d) Dependency degree under intentional attack.

attributes of NP16 are at an intermediate level among all the 47 nodes. Thus, the result of an ordinary node like NP16 is representative, which can reduce the effects of node attributes on the cascading failure

propagation process. It can be found from Fig. 10 that the number of failure nodes increases progressively with iteration rounds, and cascading failures propagate over multiple stages rather than occurring

Table 2
Correlation analysis results.

Indicators	Spearman correlation coefficient (ρ)	P
Load	-0.101	0.499
Degree centrality	-0.368	0.011
Betweenness centrality	-0.273	0.064
Dependency degree	-0.477	0.001

in one single iteration round.

To identify the failure mechanisms associated with cascading propagation at different stages, new failure nodes in each iteration round are classified into three groups according to the cause of failure, including load cascading failure, functional dependency cascading failure, and islanding failure. The classification results are summarized in Table 4. It can be seen from Table 4 that different failure mechanisms dominate at different stages of the cascading process. In the early iterations, new failure nodes are mainly associated with functional dependency cascading failure. As iterations proceed, functional dependency cascading failure, load cascading failure and islanding failure account for an increasing proportion of new failure nodes. In the later iterations, the number of new failure nodes decreases, and failures are primarily associated with load cascading failure and islanding failure until the system reaches the steady state.

SRE in every simulation iteration of NP16 under intentional attack by using four cascading failure models is shown in Fig. 11. It should be noted that the final simulation iteration round at which the iterative process terminates varies for different cascading failure models, and the maximum iteration round is set as 10. To ensure comparability of results among different models, for models whose iterative process terminates in fewer than 10 iteration rounds, it is assumed that the system remaining efficiency remains unchanged during the subsequent virtual iteration rounds until the iteration round reaches 10. It can be seen from Fig. 11 that using both the load cascading failure model and the functional dependency cascading failure model, the cascading failure process terminates in a single iteration round. Using the load and functional dependency cascading failure model in series, the cascading failure process terminates after a few iteration rounds with limited system

efficiency degradation. In contrast, using the load-functional dependency coupled cascading failure model, the cascading failure propagates over 8 iteration rounds and results in more severe system efficiency degradation. This indicates that the sustained multi-stage cascading failure process triggered by NP16 is mainly induced by the dynamic coupling between load redistribution and functional dependency degradation.

3.1.4. Sensitivity analysis of dependency coefficients

To quantitatively investigate the impacts of dependency coefficients on cascading failure propagation, a sensitivity analysis of the dependency coefficient matrix Ψ is conducted. The values in the original dependency coefficient matrix are synchronously adjusted by -30%, -20%, -10%, 0, +10%, and +20%, respectively. Fig. 12 presents the variations of SRE during cascading failure iterations under different dependency coefficient levels. It should be noted that the final simulation iteration round at which the iterative process terminates varies under different dependency coefficient levels, and the maximum iteration round in this study is set as 10. To ensure comparability of results under different dependency coefficient levels, for cases whose iterative process terminates in fewer than 10 iteration rounds, it is assumed that the system remaining efficiency remains unchanged during the subsequent virtual iteration rounds until the iteration round reaches 10.

As illustrated in Fig. 12, the dependency coefficients significantly influence the cascading failure process. The results indicate that lower dependency coefficients can effectively mitigate cascading failure

Table 3
Multiple regression analysis results.

Indicators	Beta	P
Const	0.640	2.13E-18
Load	0.104	6.52E-02
Degree centrality	-0.118	2.06E-01
Betweenness centrality	0.029	6.82E-01
Dependency degree	-0.061	3.24E-01

* Note: $R^2 = 0.245$

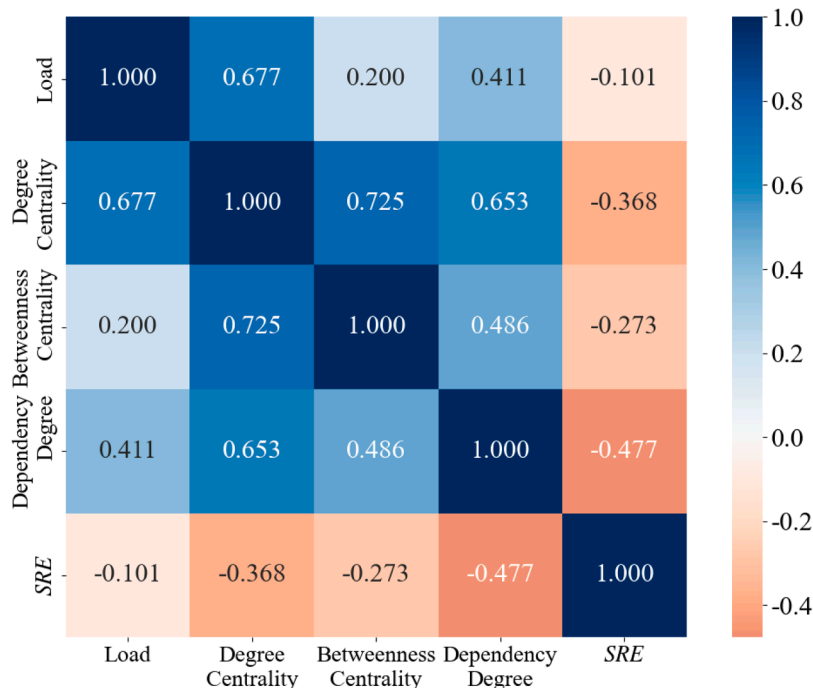


Fig. 9. Spearman correlation heatmap.

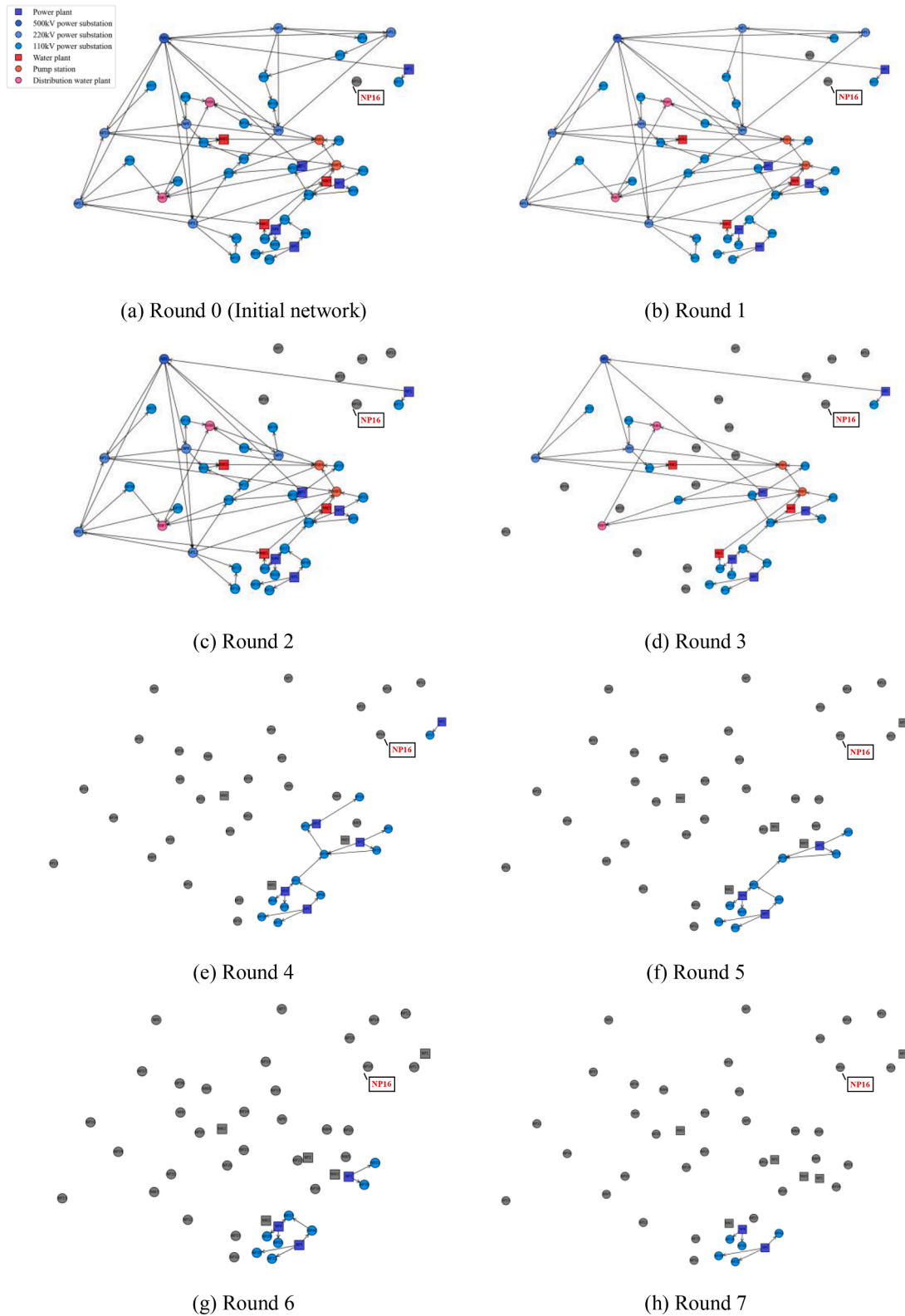


Fig. 10. Evolution of cascading failures in the power-water interdependent infrastructure network with NP16 as intentional attack node.

propagation, while stronger functional dependencies aggravate degradation of system performance. Moreover, the curves corresponding to 0, +10%, and +20% almost coincide, implying that once the dependency strength exceeds a certain threshold, the marginal impact of further enhancement becomes trivial. This sensitivity analysis further verifies the effectiveness and robustness of the proposed model in characterizing

functional dependency effects.

3.2. Network vulnerability analyses under regional attacks

To analyze network vulnerability under regional disturbances, regional attack scenarios are simulated by selecting representative

Table 4

New failure nodes and corresponding failure causes in each iteration under NP16 as initial failure node.

Iteration round	Number of new failure nodes	Failure causes		
		Load redistribution	Functional dependency	Islanding
0	1	/	/	/
1	1	/	NP15	/
2	4	NP11	NP7, NP18	NP14
3	10	/	NP9, NP12, NP13, NP19, NP23, NP24	NP32, NP33, NP34, NP35
4	14	NP6, NP8, NP26	NP10, NP25, NP36, NP37, NW1, NW2, NW3, NW4, NW5, NW6, NW7	
5	5	NP1, NP2, NP22	/	NP17, NP20
6	1	NP38	/	/
7	4	NP3, NP27, NP39	/	NP21
8	0	/	/	/

central attack nodes with different dependency degrees. System remaining efficiencies under regional attacks changing with attack intensity and attack range are shown in Fig. 13 and Fig. 14, respectively.

When the attack range is fixed as 2, the variation of *SRE* with attack intensity is shown in Fig. 13. It is evident from Fig. 13 that *SRE* decreases nonlinearly as attack intensity increases. For low attack intensities, the reduction in system functionality is limited. This is because when attack intensity is below critical level, system redundancy can mitigate the localized impacts and limit cascading propagation. As attack intensity increases beyond a certain level, *SRE* drops rapidly. This is because when attack intensity is beyond critical level, the combined effects of load redistribution and functional dependency degradation drive cascading failures to transition rapidly from localized impacts to large-scale system collapse, resulting in drastic system functionality loss. Moreover, the larger the dependency degree of the central attack node is, the more significant the system efficiency degradation is. With further increases in attack intensity, *SRE* tends to stabilize at a certain level.

When the attack intensity is fixed as 0.2, the variation of *SRE* with attack range is shown in Fig. 14. According to the results in Fig. 13, when the attack intensity exceeds 0.2, *SRE* decreases rapidly and gradually approaches a relatively stable level. Therefore, the attack intensity is set as 0.2 in Fig. 14 to better illustrate the influence of attack range on system vulnerability under a representative disturbance condition. It can be seen from Fig. 14 that increasing the attack range leads to progressive system functionality degradation. For central attack nodes with higher dependency degree, a relatively small attack range results in a marked reduction in *SRE*, whereas for central attack nodes with lower dependency degree, a larger attack range is required to produce a comparable decrease in system functionality.

4. Conclusions and future work

This study develops a load-functional dependency coupled cascading failure model to characterize progressive failure processes in power-water interdependent infrastructure networks under complex disturbances. By integrating load redistribution and execution-performance-based functionality degradation within a unified dynamic framework, the proposed model captures bidirectional feedback mechanisms that are neglected in previous studies on vulnerability analyses of interdependent critical infrastructure networks. The proposed model is applied to analyze vulnerability of a county-scale power-water interdependent infrastructure network in Zibo, China. The following conclusions can be drawn:

- (1) Compared with load cascading failure model and functional dependency cascading failure model, the load-functional dependency coupled model leads to faster failure propagation and more extensive system functionality degradation. Neglecting the coupling of load dynamics and functional dependency will underestimate system vulnerability in power-water interdependent infrastructure networks.
- (2) From perspective of the load-functional dependency coupled model, vulnerability of power-water interdependent infrastructure networks arises from dynamic interactions among load redistribution, functional dependency propagation, rather than

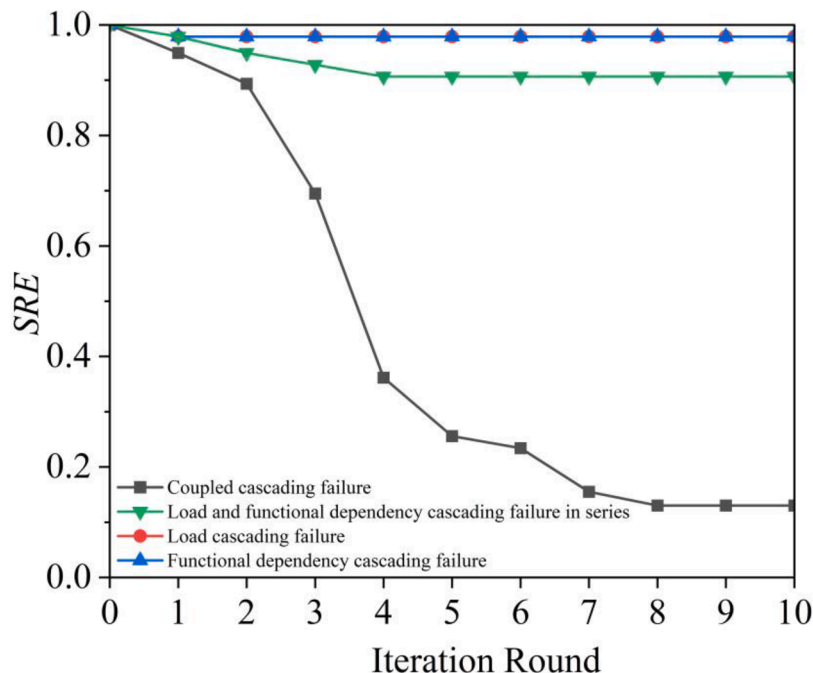


Fig. 11. System remaining efficiencies in every simulation iteration of NP16 under intentional attack.

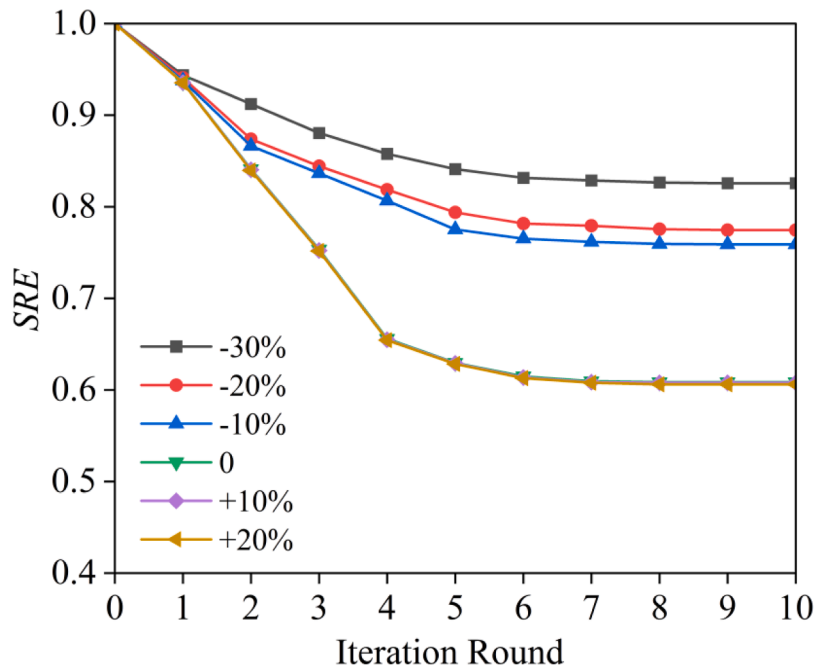


Fig. 12. Sensitivity analysis of dependency coefficients on SRE during cascading failure iterations.

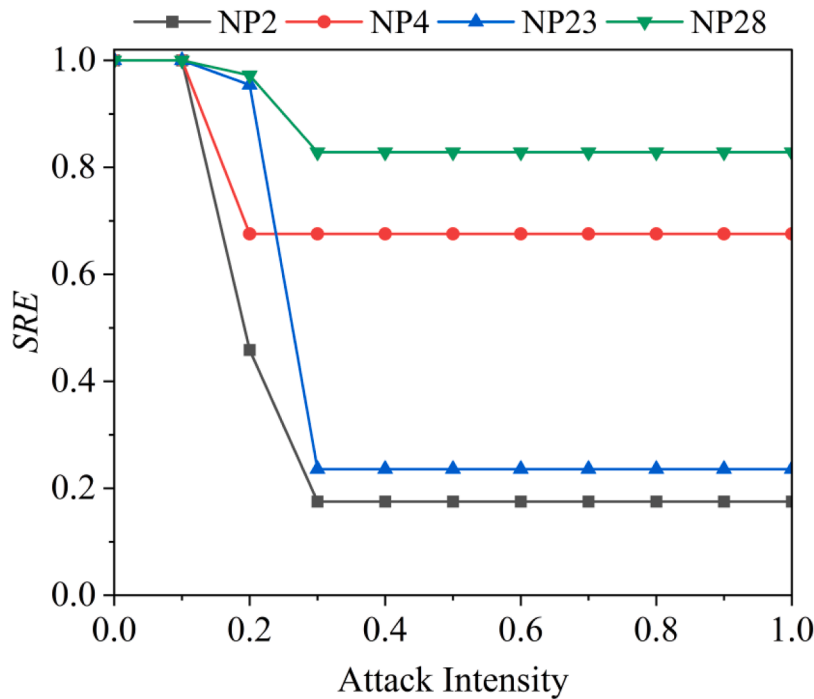


Fig. 13. SRE changing with attack intensity where attack range is 2.

node attributes including node load, degree centrality, betweenness centrality, and dependency degree.

- (3) When attack intensity is below critical level, system redundancy can mitigate the localized impacts and limit cascading propagation. As attack intensity increases beyond a certain level, system efficiency drops rapidly. Moreover, the larger the dependency degree of the central attack node is, the more significant the system efficiency degradation is.

However, several limitations still exist in this study. Firstly, this study only considers cascading failures induced by node failures, while

cascading failures caused by physical link failures, such as transmission line failures and water pipe failures, are not considered. In real-world infrastructure systems, link failures may also significantly affect network functionality and trigger cascading propagation processes. Secondly, compared with physics-based infrastructure network models considering hydraulic and power flow mechanisms, the adopted topology-based models may underestimate the vulnerability of real-world infrastructure networks because the impacts of physical operating constraints and flow redistribution processes after failures are not explicitly considered. Therefore, integrating physical operating mechanisms and flow models into coupled cascading failure analyses can be

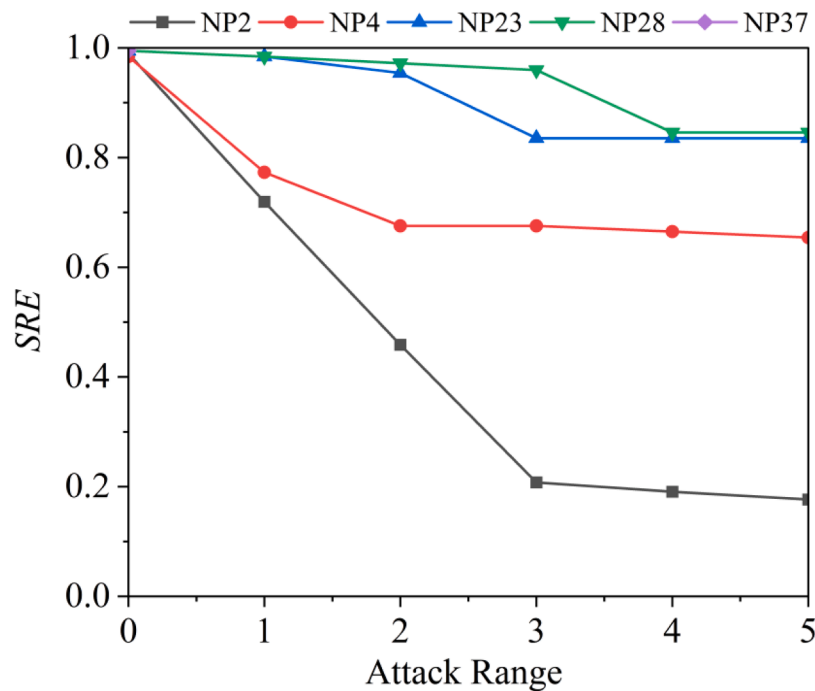


Fig. 14. SRE changing with attack range where attack intensity is 0.2.

further explored in future work. Thirdly, only the unidirectional inter-layer dependency of water infrastructures on power infrastructures is considered in this study. Bidirectional interdependencies widely exist in real-world power-water infrastructure systems, where the infrastructure network should be modelled considering loops. Cascading failure propagation behaviors under bidirectional interdependency conditions can be further investigated to improve the applicability of the proposed model.

CRedit authorship contribution statement

Xinnan Liu: Methodology, Conceptualization. **Jialu Li:** Writing – original draft, Formal analysis, Data curation. **Zhanling Li:** Visualization, Software. **Yingbo Ji:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.res.2026.113092](https://doi.org/10.1016/j.res.2026.113092).

Data availability

Data will be made available on request.

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